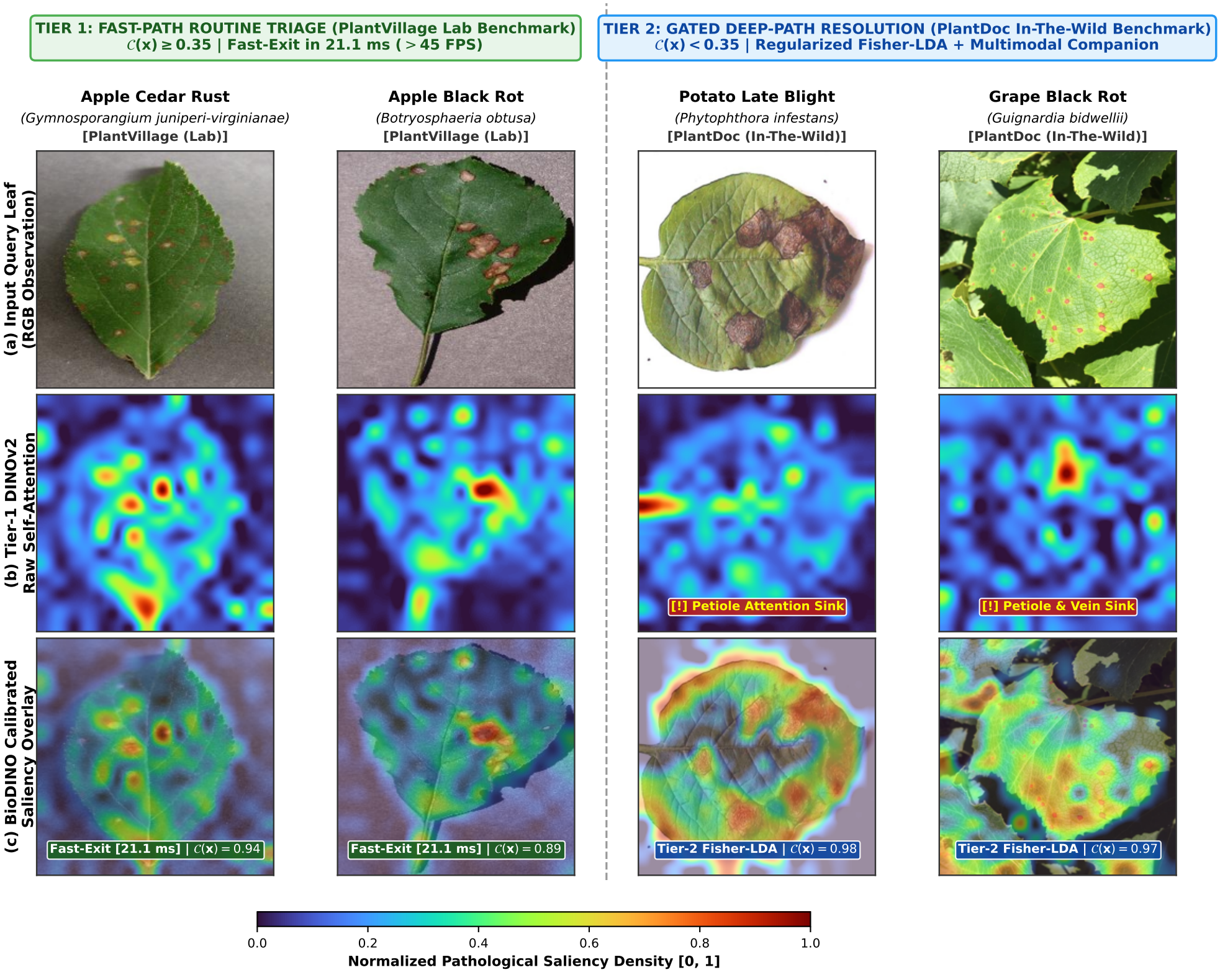

TIER 1: FAST-PATH ROUTINE TRIAGE (PlantVillage Lab Benchmark)
$\mathcal{C}(\mathbf{x}) \geq 0.35$ | Fast-Exit in 21.1 ms (> 45 FPS)
TIER 2: GATED DEEP-PATH RESOLUTION (PlantDoc In-The-Wild Benchmark)
$\mathcal{C}(\mathbf{x}) < 0.35$ | Regularized Fisher-LDA + Multimodal Companion
Apple Cedar Rust
(Gymnosporangium juniperi-virginianae)
[PlantVillage (Lab)]
Apple Black Rot
(Botryosphaeria obtusa)
[PlantVillage (Lab)]
Potato Late Blight
(Phytophthora infestans)
[PlantDoc (In-The-Wild)]
Grape Black Rot
(Guignardia bidwellii)
[PlantDoc (In-The-Wild)]
(a) Input Query Leaf (RGB Observation)
(b) Tier-1 DINOv2 Raw Self-Attention
(c) BioDINO Calibrated Saliency Overlay
[!] Petiole Attention Sink
[!] Petiole & Vein Sink
Fast-Exit [21.1 ms] | $\mathcal{C}(\mathbf{x}) = 0.94$
Fast-Exit [21.1 ms] | $\mathcal{C}(\mathbf{x}) = 0.89$
Tier-2 Fisher-LDA | $\mathcal{C}(\mathbf{x}) = 0.98$
Tier-2 Fisher-LDA | $\mathcal{C}(\mathbf{x}) = 0.97$
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Pathological Saliency Density [0, 1]